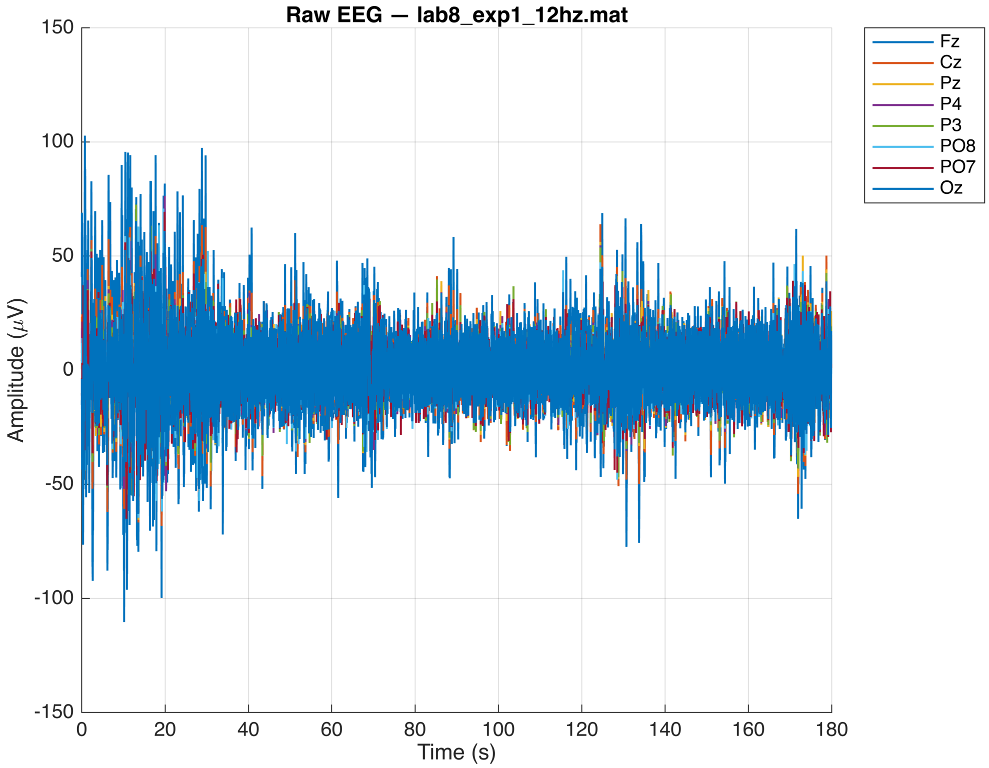
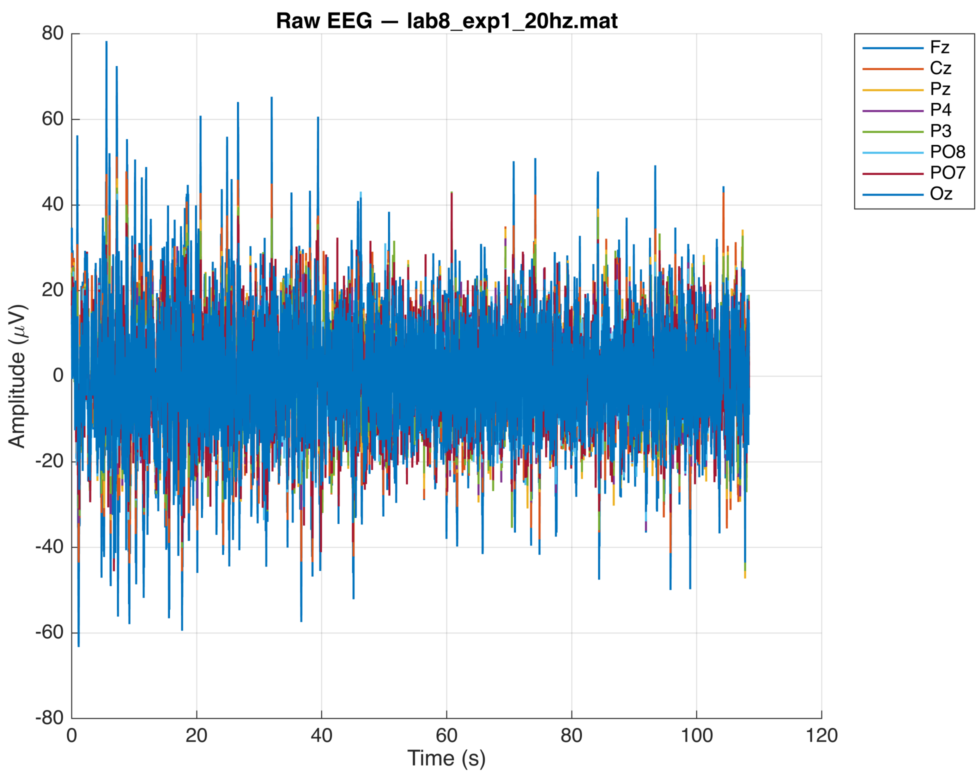
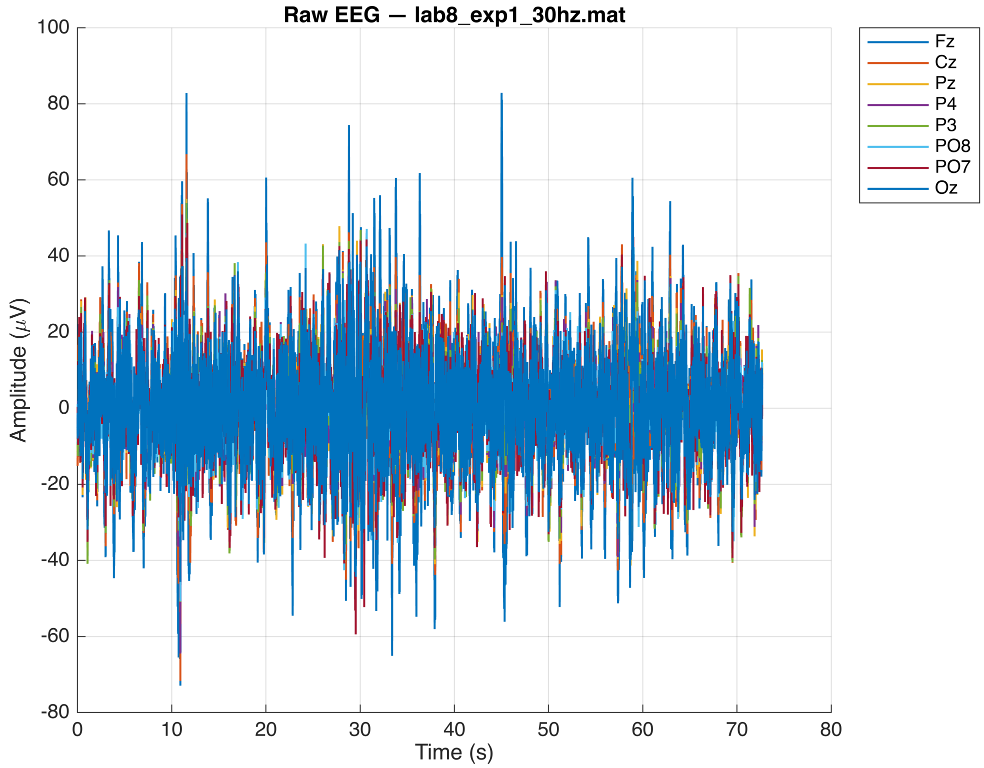
Plot the raw waveforms.

```
clear; close all; clc;

fs = 256; %the sampling rate we're using
remove_sec = 10;
Nremove = fs * remove_sec;

%% plotting just exp1, our 12 20 and 30 hz recordings.

exp1_files = { ...
    'lab8_exp1_12hz.mat', ...
    'lab8_exp1_20hz.mat', ...
    'lab8_exp1_30hz.mat'};

chanNames = {'Fz','Cz','Pz','P4','P3','PO8','PO7','Oz'};

for f = 1:numel(exp1_files)
    S = load(exp1_files{f});
    fn = fieldnames(S);
    pre = S.(fn{1}); % this is a 9 by n matrix

    t   = pre(1,:);
    EEG = pre(2:9,:);  % our 8 channels w the first 1 = time

    % removing the first 10s
    t   = t(Nremove+1:end);
    t   = t - t(1);
    EEG = EEG(:, Nremove+1:end);

    % just to detrend, remove dc drift
    for ch = 1:8
        EEG(ch,:) = detrend(EEG(ch,:), 'constant');
    end

    figure('Color','w');
    hold on;
    for ch = 1:8
        plot(t, EEG(ch,:), 'LineWidth', 1);
    end
    hold off;

    xlabel('Time (s)');
    ylabel('Amplitude (\muV)');
    title(['Raw EEG - ' strrep(exp1_files{f}, '_','\_')]);
    legend(chanNames, 'Location','bestoutside');
    grid on;
end
```

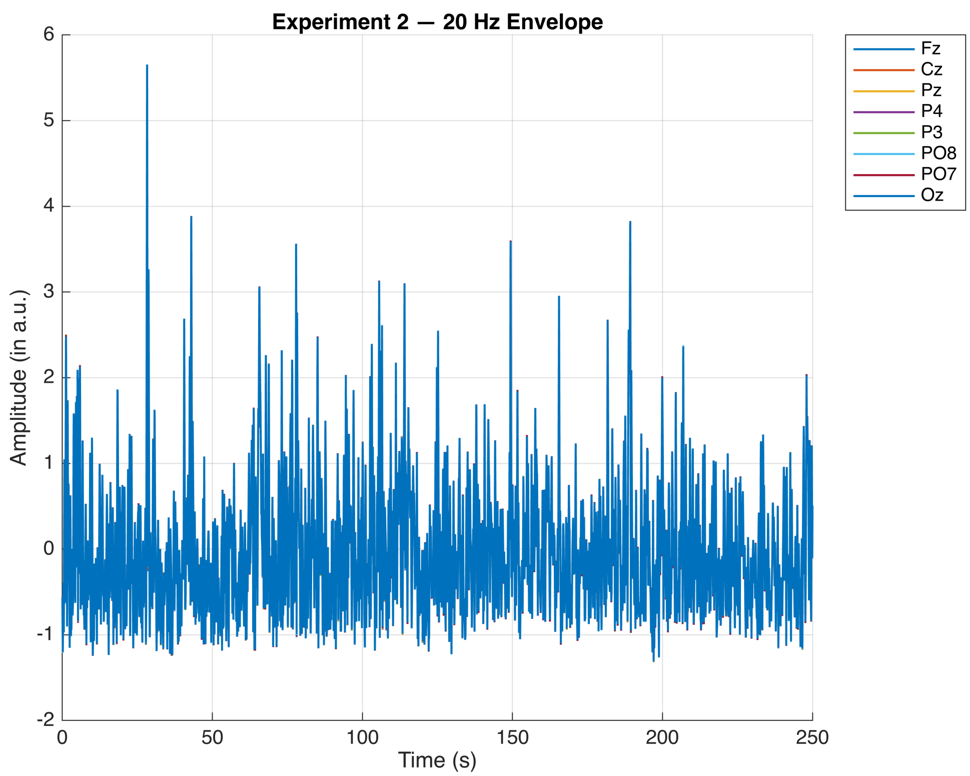

Raw EEG — lab8_exp1_12hz.mat
Amplitude (μV)
Time (s)
150
100
50
0
-50
-100
-150
0
20
40
60
80
100
120
140
160
180
Fz
Cz
Pz
P4
P3
PO8
PO7
Oz


Raw EEG — lab8_exp1_20hz.mat
Amplitude (μV)
Time (s)
80
60
40
20
0
-20
-40
-60
-80
0
20
40
60
80
100
120
Fz
Cz
Pz
P4
P3
PO8
PO7
Oz



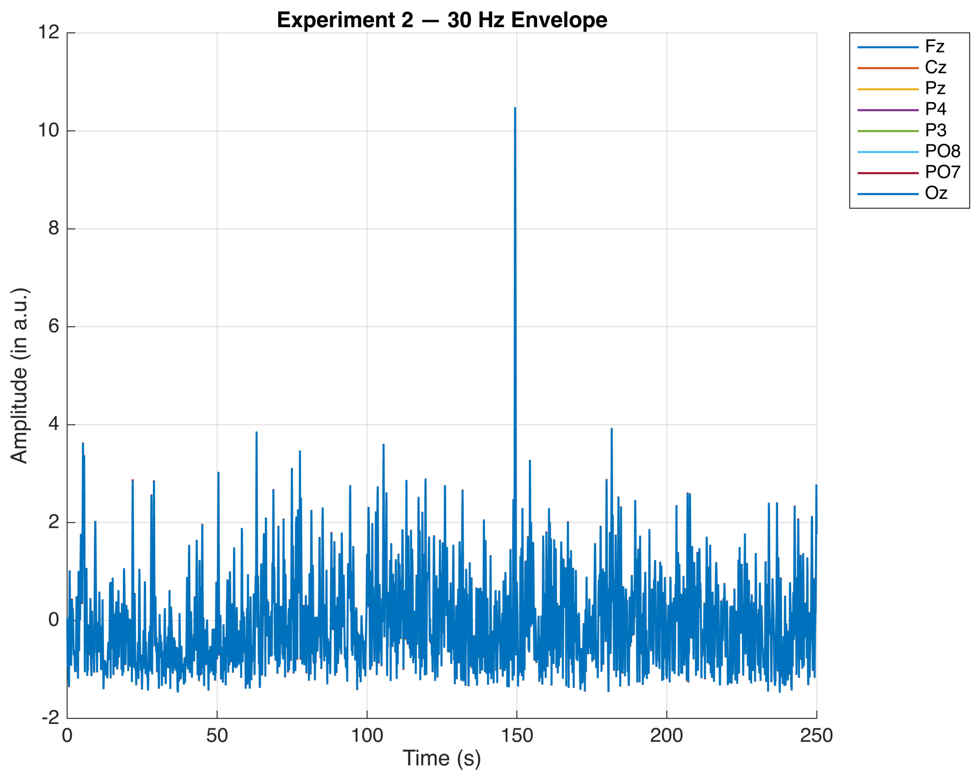

```
%experiment 2, now looking at the envelope data
%just making sure that it makes sense.

file_exp2 = 'lab8_exp2_individualChannels_trial3.mat';
S2 = load(file_exp2);
y = S2.y;% this var 17x66561 M

% the data structure:
%row 1      = time
%rows 2–9   = 20 Hz envelope
%rows 10–17 = 30 Hz envelope

t      = y(1,:);
env20  = y(2:9,:);
env30  = y(10:17,:);

%again remove first 10s
t     = t(Nremove+1:end);
t     = t - t(1);
env20 = env20(:, Nremove+1:end);
env30 = env30(:, Nremove+1:end);

% then detrend the data
for ch = 1:8
```

```
    env20(ch,:) = detrend(env20(ch,:), 'constant');
    env30(ch,:) = detrend(env30(ch,:), 'constant');
end

%% 20Hz same plotting style as exp1
figure('Color','w');
hold on;
for ch = 1:8
    plot(t, env20(ch,:), 'LineWidth', 1);
end
hold off;

xlabel('Time (s)');
ylabel('Amplitude (in a.u.)');
title('Experiment 2 — 20 Hz Envelope');
legend(chanNames, 'Location','bestoutside');
grid on;
```




```
%% 30hz
figure('Color','w');
hold on;
for ch = 1:8
    plot(t, env30(ch,:), 'LineWidth', 1);
end
hold off;
```

```
xlabel('Time (s)');
ylabel('Amplitude (in a.u.)');
title('Experiment 2 — 30 Hz Envelope');
legend(chanNames, 'Location','bestoutside');
grid on;
```

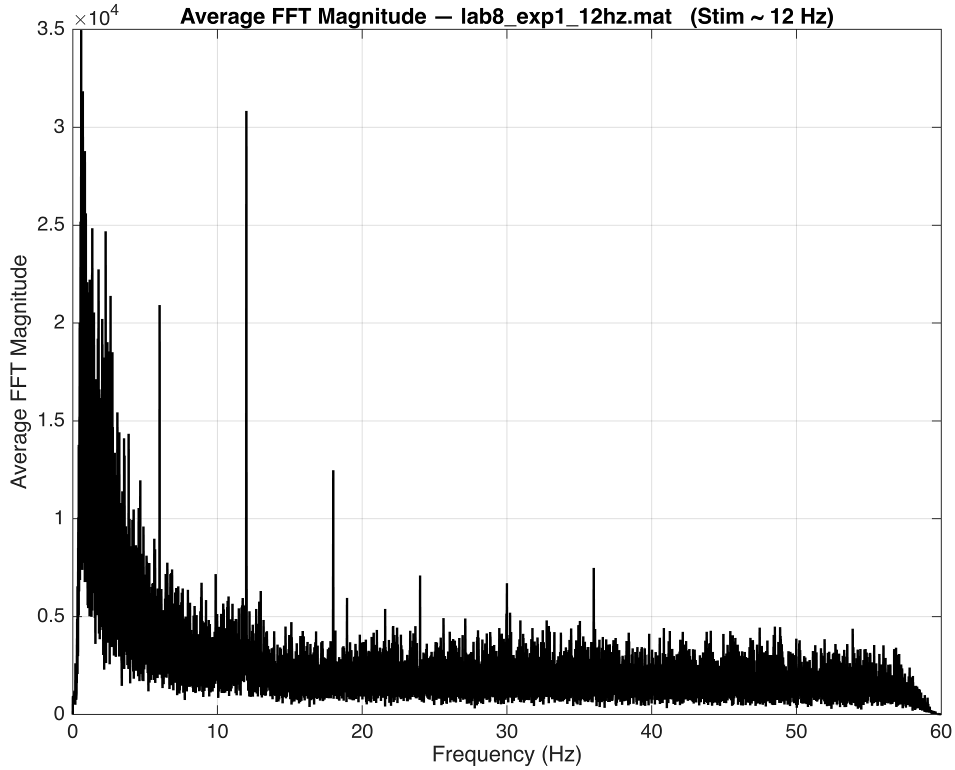


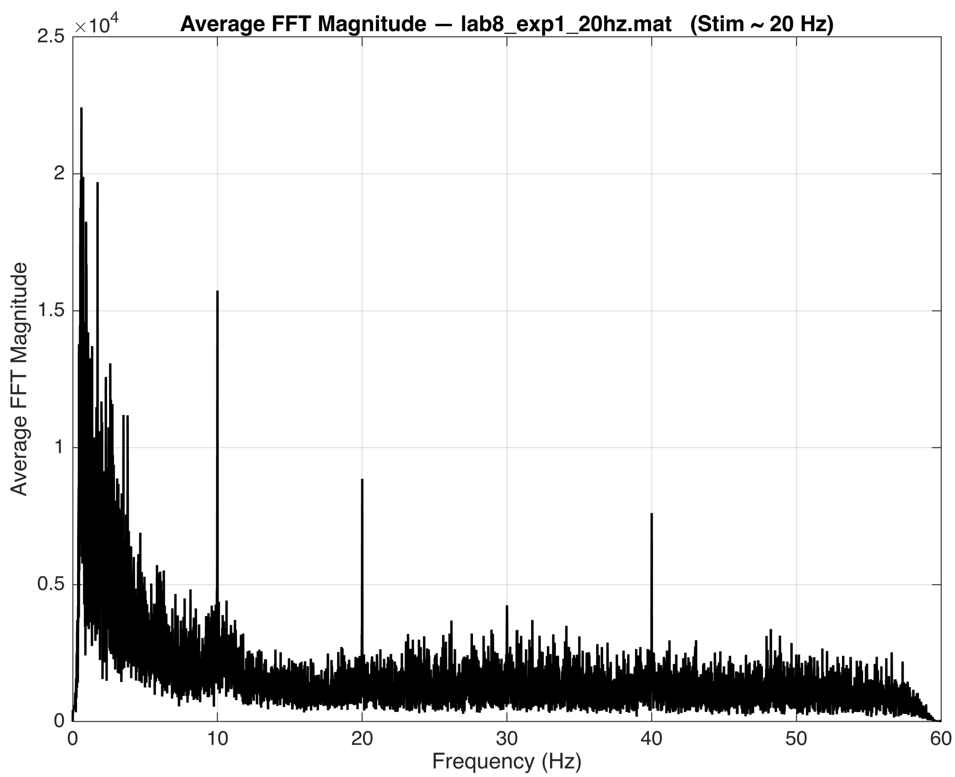

```

```

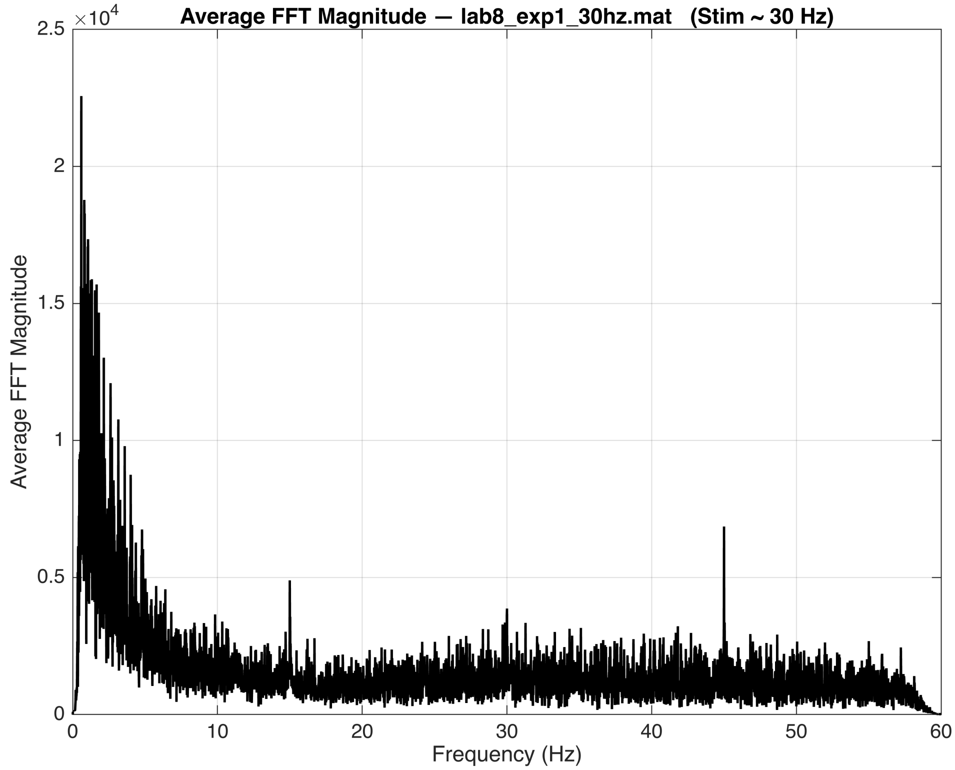
Load the EEG signal in MATLAB. Start by taking the Fourier transform of each channel and averaging the FFT magnitude over all channels. Plot the average magnitude. (1 pt)

```
fs = 256;
remove_sec = 10;
Nremove = fs * remove_sec;

exp1_files = { ...
    'lab8_exp1_12hz.mat', ...
    'lab8_exp1_20hz.mat', ...
    'lab8_exp1_30hz.mat'};

stim_freqs = [12 20 30]; % for labeling plots
chanNames   = {'Fz','Cz','Pz','P4','P3','PO8','PO7','Oz'};

for f = 1:numel(exp1_files)
    %almost all of the same preprocessing steps
    S = load(exp1_files{f});
```

```
    fn = fieldnames(S);
    pre = S.(fn{1}); % 9 x n

    t   = pre(1,:);
    EEG = pre(2:9,:); %8

    t   = t(Nremove+1:end);
    EEG = EEG(:, Nremove+1:end);

    for ch = 1:8
        EEG(ch,:) = detrend(EEG(ch,:), 'constant');
    end

    % start the FFT
    L = size(EEG,2);
    freqs = (0:L-1)*(fs/L);

    % for each of our channels
    fft_all = fft(EEG, [], 2);  %8 x L
    mag_all = abs(fft_all); % the magnitude

    % then well take the avg magnitude across all the channels
    avg_mag = mean(mag_all, 1);

    % then plot
    figure('Color','w');
    plot(freqs, avg_mag, 'k', 'LineWidth', 1.2);
    xlim([0 60]); % just because this is the band of interest
    xlabel('Frequency (Hz)');
    ylabel('Average FFT Magnitude');
    title(['Average FFT Magnitude - ' strrep(exp1_files{f}, '_','\_') ...
           '   (Stim ~ ' num2str(stim_freqs(f)) ' Hz)']);
    grid on;

end
```


Average FFT Magnitude — lab8_exp1_12hz.mat (Stim ~ 12 Hz)
$\times10^4$
3.5
3
2.5
2
1.5
1
0.5
0
Average FFT Magnitude
0
10
20
30
40
50
60
Frequency (Hz)


Average FFT Magnitude — lab8_exp1_20hz.mat (Stim ~ 20 Hz)
$\times10^4$
2.5
2
1.5
1
0.5
0
Average FFT Magnitude
0
10
20
30
40
50
60
Frequency (Hz)



Question 1. If the length of the FFT is L, given the sampling frequency of your EEG signal (256Hz), which sample in FFT corresponds to each frame rate? Mark that frequency on the plots (you can zoom in to increase the resolution). (2 pts)

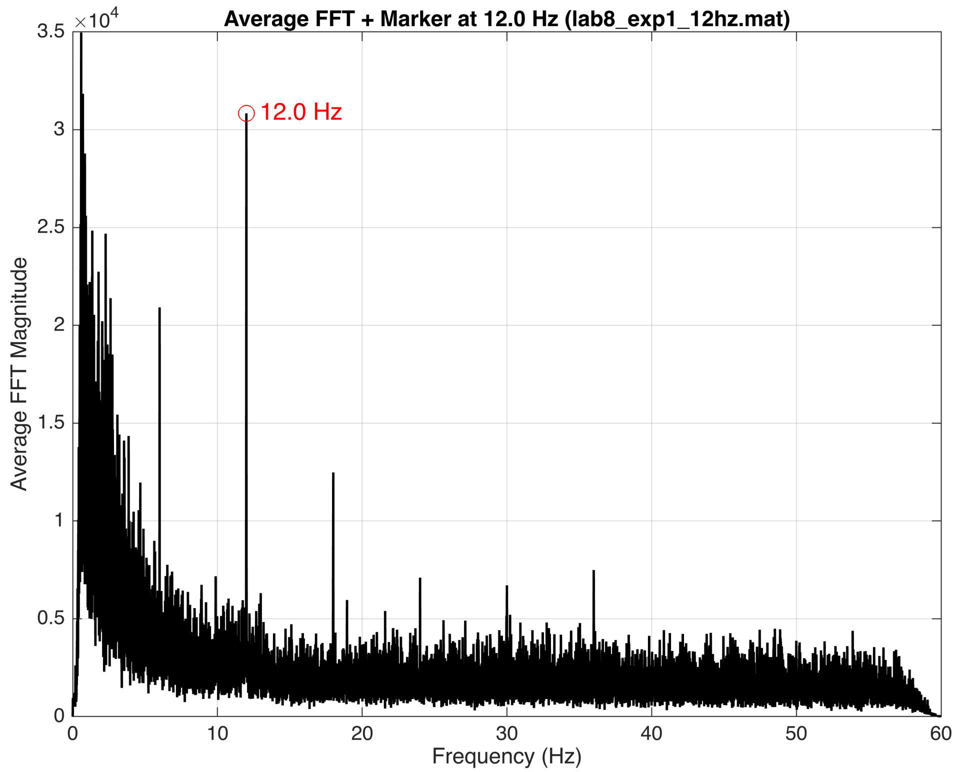
Next, display all channels at the modulator frequency.

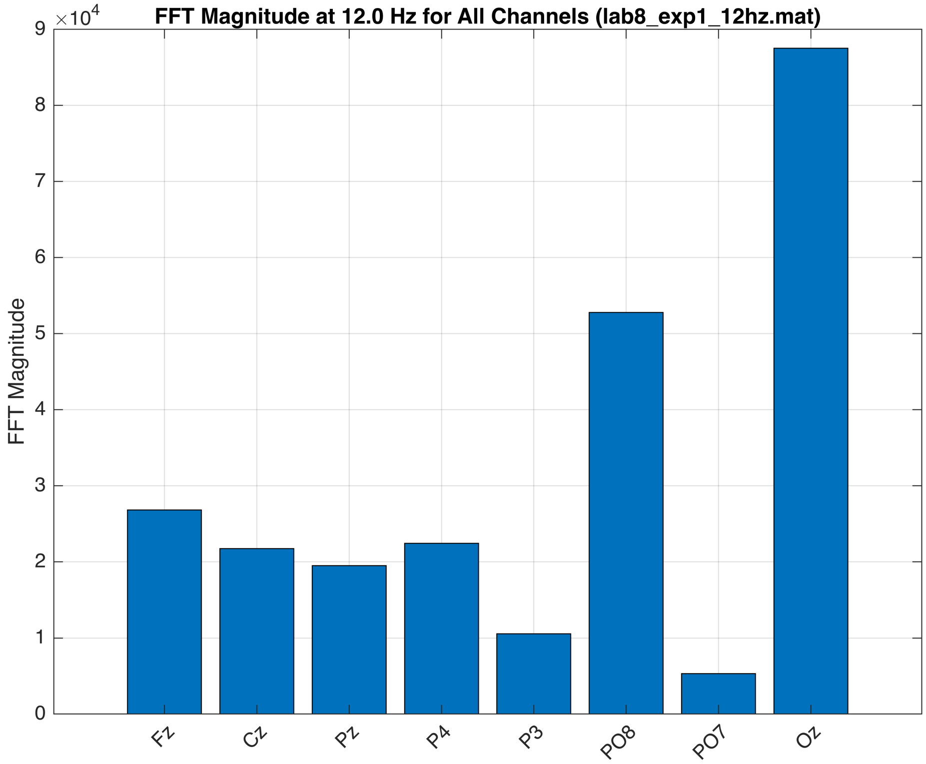
```
fs = 256;
remove_sec = 10;
Nremove = fs * remove_sec;

files = {'lab8_exp1_12hz.mat', ...
         'lab8_exp1_20hz.mat', ...
         'lab8_exp1_30hz.mat'};

stim_freqs = [12 20 30];
chanNames = {'Fz','Cz','Pz','P4','P3','PO8','PO7','Oz'};

%we are going to do it for every file + many sanity checks to make sure
for i = 1:numel(files)

    fprintf('File: %s\n', files{i});
    fprintf('Stimulus frequency: %.1f Hz\n', stim_freqs(i));

    %just to load and trim file
    S = load(files{i});
```

```
fn = fieldnames(S);
pre = S.(fn{1});

t   = pre(1,:);
EEG = pre(2:9,:);
EEG = EEG(:, Nremove+1:end);

for ch = 1:8
    EEG(ch,:) = detrend(EEG(ch,:), 'constant');
end

%fft again
L = size(EEG,2);
df = fs / L;

fft_all = fft(EEG, [], 2);
mag_all = abs(fft_all);
avg_mag = mean(mag_all, 1);

%indentify each of the bins
f0 = stim_freqs(i);
bin = round(f0 / df) + 1;

fprintf('FFT length L = %d samples\n', L);
fprintf('Frequency resolution df = %.6f Hz/bin\n', df);
fprintf('FFT bin for %.1f Hz = %d\n', f0, bin);

% should just have our sanity checks
f_reconstructed = (bin - 1) * df;
fprintf('Sanity check: bin %d → %.6f Hz (should be %.1f Hz)\n\n', ...
        bin, f_reconstructed, f0);

%plotting the avg fft with the marker included for visibility
freqs = (0:L-1)*(fs/L);

figure('Color','w');
plot(freqs, avg_mag, 'k', 'LineWidth', 1.2); hold on;
plot(freqs(bin), avg_mag(bin), 'ro', 'MarkerSize', 8);
text(freqs(bin), avg_mag(bin), ...
     sprintf('  %.1f Hz', f0), 'Color','r','FontSize',12);

xlim([0 60]);
xlabel('Frequency (Hz)');
ylabel('Average FFT Magnitude');
title(sprintf('Average FFT + Marker at %.1f Hz (%s)', ...
      f0, strrep(files{i}, '_','\_')));
grid on;

% just dsp each channel at f0
mag_at_f0 = mag_all(:, bin);%8x1
```

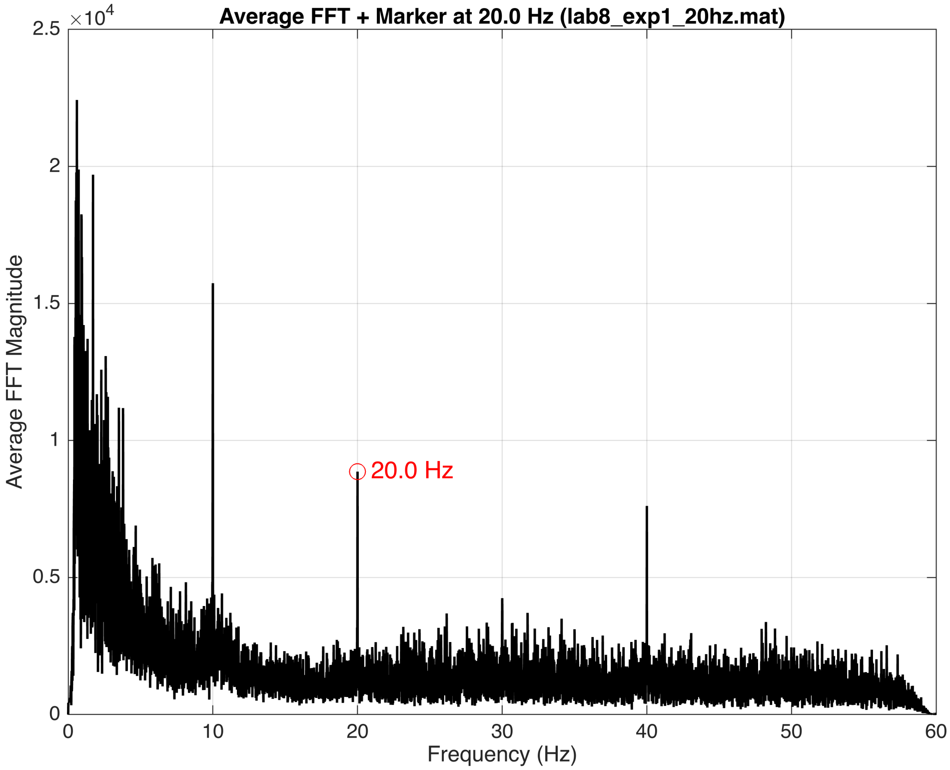
```
    figure('Color','w');
    bar(mag_at_f0);
    set(gca,'XTickLabel',chanNames,'XTickLabelRotation',45);

    ylabel('FFT Magnitude');
    title(sprintf('FFT Magnitude at %.1f Hz for All Channels (%s)', ...
          f0, strrep(files{i}, '_','\_')));
    grid on;

end
```

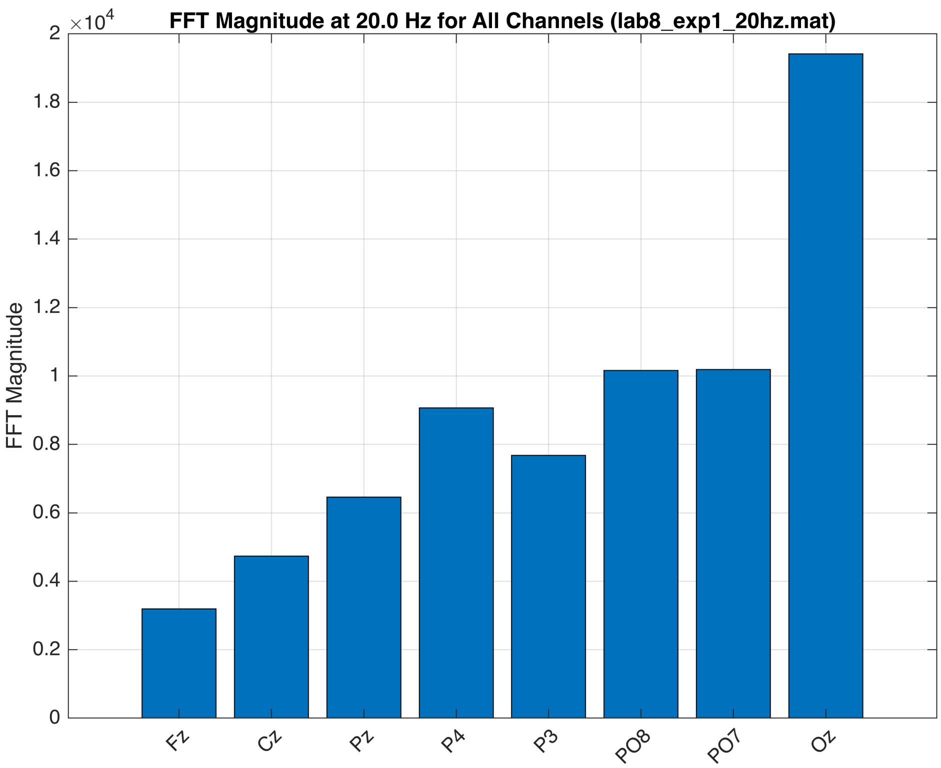
```
File: lab8_exp1_12hz.mat
Stimulus frequency: 12.0 Hz
FFT length L = 46062 samples
Frequency resolution df = 0.005558 Hz/bin
FFT bin for 12.0 Hz = 2160
Sanity check: bin 2160 → 11.999132 Hz (should be 12.0 Hz)
```





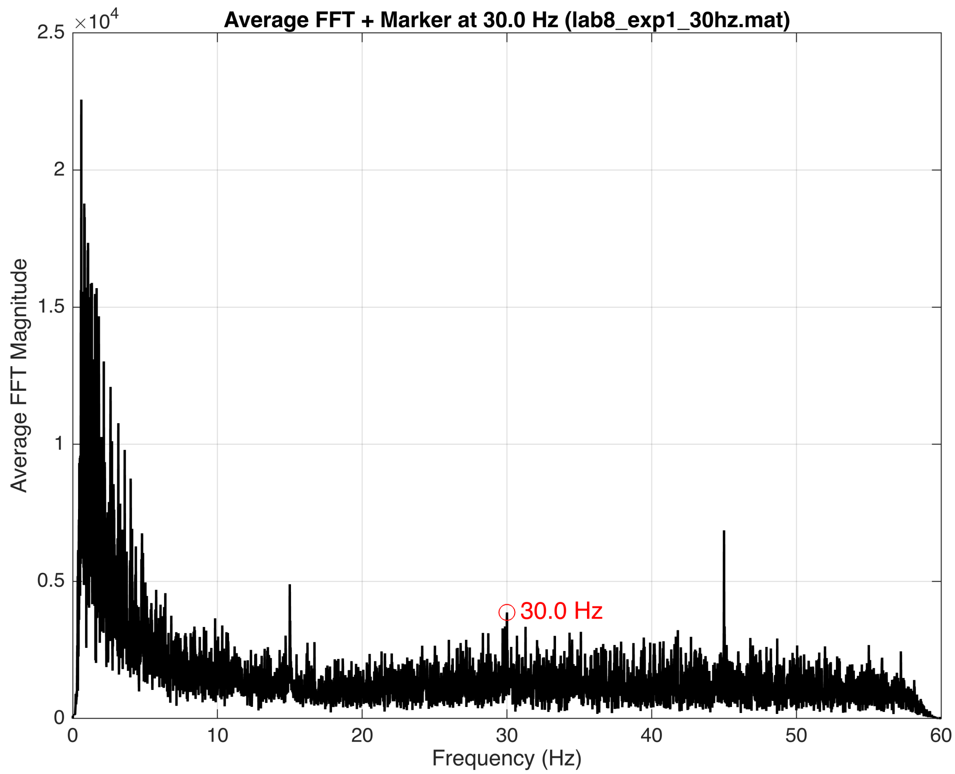

```
File: lab8_exp1_20hz.mat
Stimulus frequency: 20.0 Hz
FFT length L = 27748 samples
Frequency resolution df = 0.009226 Hz/bin
FFT bin for 20.0 Hz = 2169
Sanity check: bin 2169 → 20.001730 Hz (should be 20.0 Hz)
```

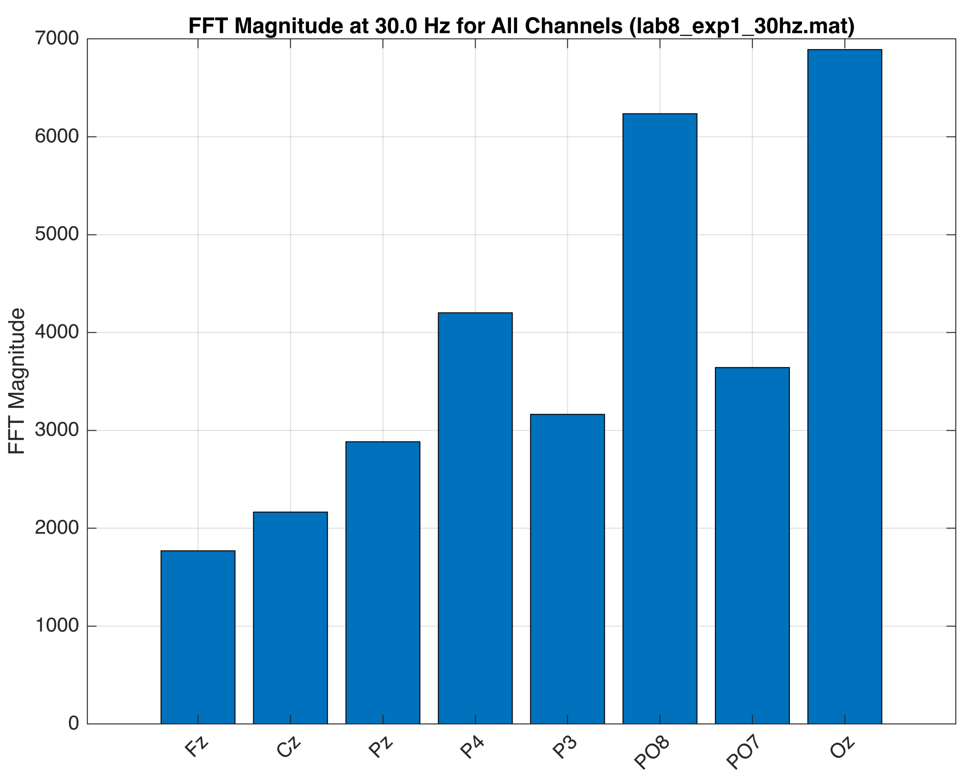





```
File: lab8_exp1_30hz.mat
Stimulus frequency: 30.0 Hz
FFT length L = 18608 samples
Frequency resolution df = 0.013758 Hz/bin
```

```
FFT bin for 30.0 Hz = 2182
Sanity check: bin 2182 → 30.005159 Hz (should be 30.0 Hz)
```





Question 2. Create scalp maps showing the average magnitude of the SSVEP signal over all channels for all three stimuli used. Which stimulus generated the maximum response? Which scalp areas had the strongest SSVEP? (3 pts)

```
%have to add this path manually to make sure.
addpath('/Users/alexanderspeer/Desktop/BCI lab/eeglab14_1_2b');
eeglab;
```

```
Warning: Adding folders named 'resources' to the path is not supported: /Users/alexanderspeer/
Desktop/BCI lab/eeglab14_1_2b/functions/resources
Warning: The value of local variables may have been changed to match the globals.  Future versions
of MATLAB will require that you declare a variable to be global before you use that variable.
Warning: colordef will be removed in a future release.
eeglab: options file is ~/eeg_options.m
EEGLAB: error while adding plugin "eegplugin_dipfit"
   Error using uimenu
First argument must be a valid parent, such as a Figure or Panel object.
EEGLAB: error while adding plugin "eegplugin_firfilt"
   Error using uimenu
First argument must be a valid parent, such as a Figure or Panel object.
```

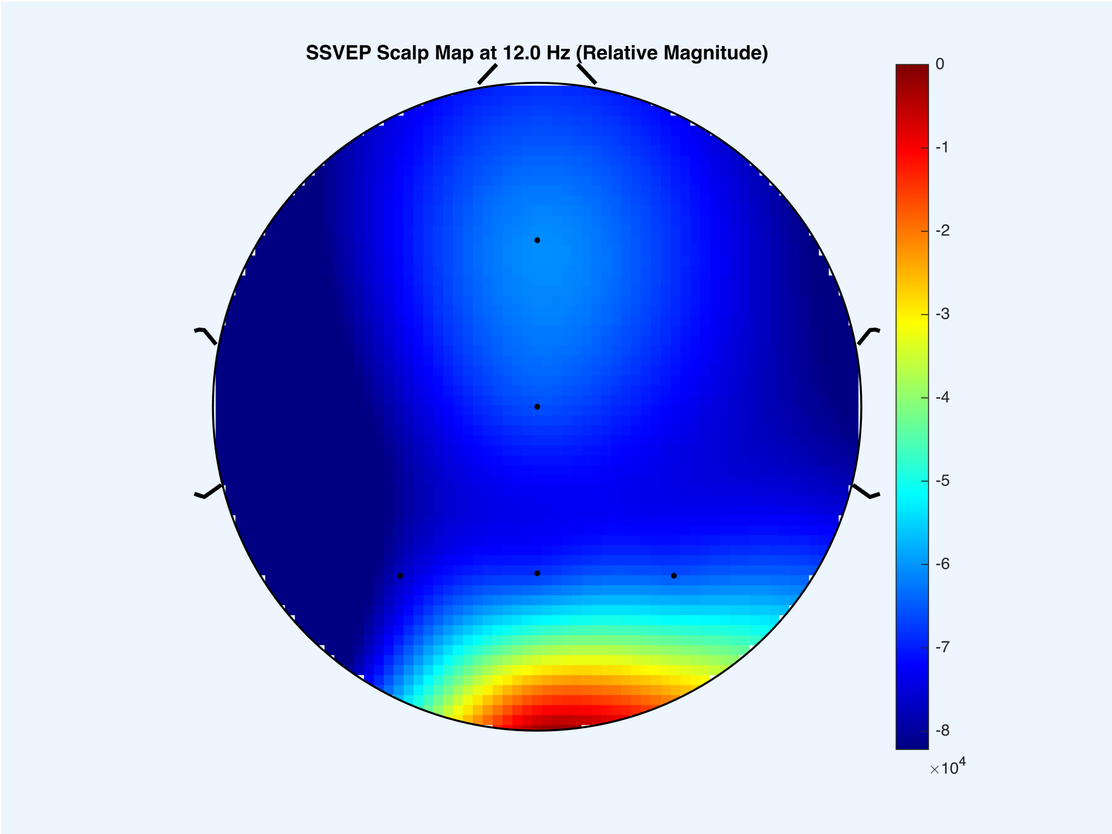
```
%then run the eeglab cmd in the CLI

fs = 256;
remove_sec = 10;
Nremove = fs * remove_sec;

files = {'lab8_exp1_12hz.mat', ...
         'lab8_exp1_20hz.mat', ...
         'lab8_exp1_30hz.mat'};

stim_freqs = [12 20 30];
chanNames  = {'Fz','Cz','Pz','P4','P3','PO8','PO7','Oz'};

% load the channel locations from the loc file
locs_path = 'BCI.locs';
chanlocs_full = readlocs(locs_path);
```

```
readlocs(): 'loc' format assumed from file extension
```

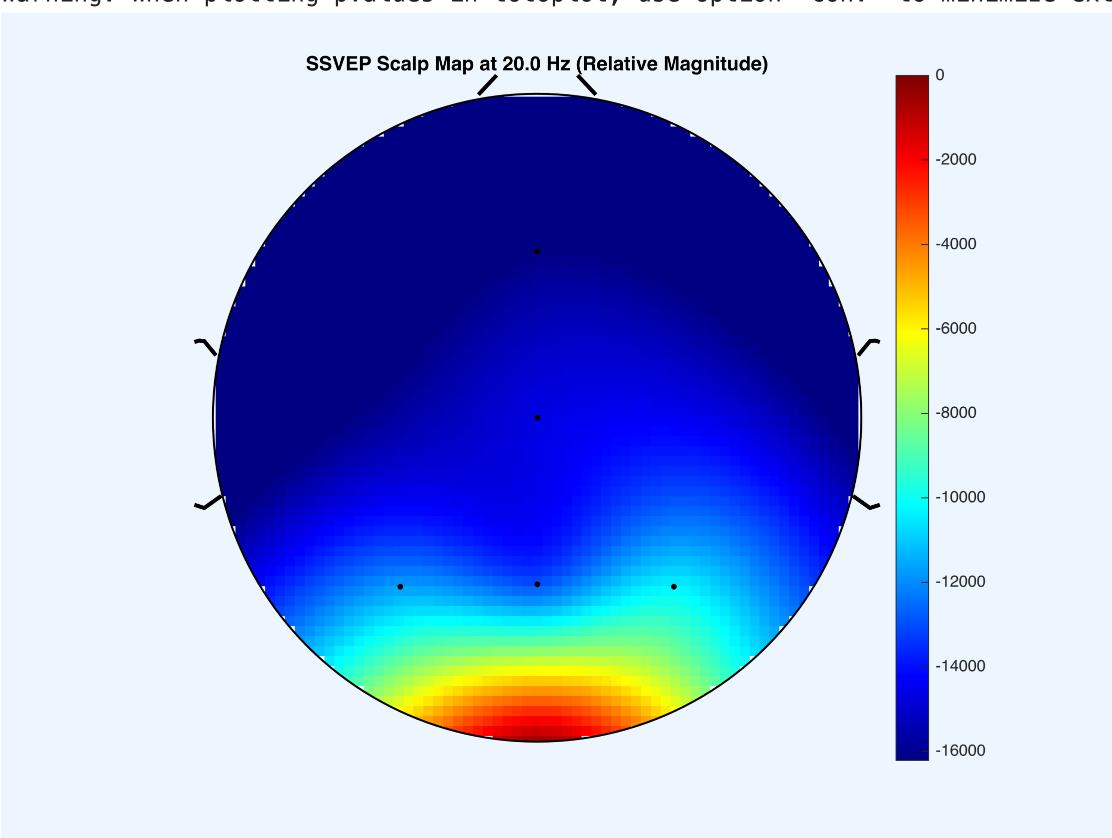
```

% have to make sure indices correspond to the actually channels
%that were used in the exp.
% Fz, Cz, Pz, P4, P3, PO8, PO7, Oz
selLocIdx = [1 6 11 12 13 14 15 16];

chanlocs = chanlocs_full(selLocIdx);

% process each stim. condition
magnitudes_all = zeros(3,8);   % later comparision.

for i = 1:numel(files)
```

```
    S = load(files{i});
    fn = fieldnames(S);
    pre = S.(fn{1}); % 9 x N Matrix

    t   = pre(1,:);
    EEG = pre(2:9,:);
    EEG = EEG(:, Nremove+1:end);

    for ch = 1:8
        EEG(ch,:) = detrend(EEG(ch,:), 'constant');
    end

    %fft
    L = size(EEG,2);
    df = fs / L;

    fft_all = fft(EEG, [], 2);
    mag_all = abs(fft_all);

    %the bin for f0
    f0 = stim_freqs(i);
    bin = round(f0 / df) + 1;

    %magnitude for each channel
    mags = mag_all(:, bin);
    magnitudes_all(i,:) = mags;

    % normalize for our visualiztion, with the max channel at 0 dB
    vals = mags - max(mags);

    % the topoplot
    figure('Color','w');
    topoplot(vals, chanlocs, ...
        'electrodes','on', ...
        'numcontour', 6, ...
        'style','map', ...
        'plotrad', 0.5);

    caxis([min(vals) 0]); %?
    colorbar;
    title(sprintf('SSVEP Scalp Map at %.1f Hz (Relative Magnitude)', f0));

    %just print the strongest channel we have.
    [mx, ix] = max(mags);
    fprintf('%s: %.1f Hz strongest = %s (Magnitude %.2f)\n', ...
            files{i}, f0, chanNames{ix}, mx);
end
```

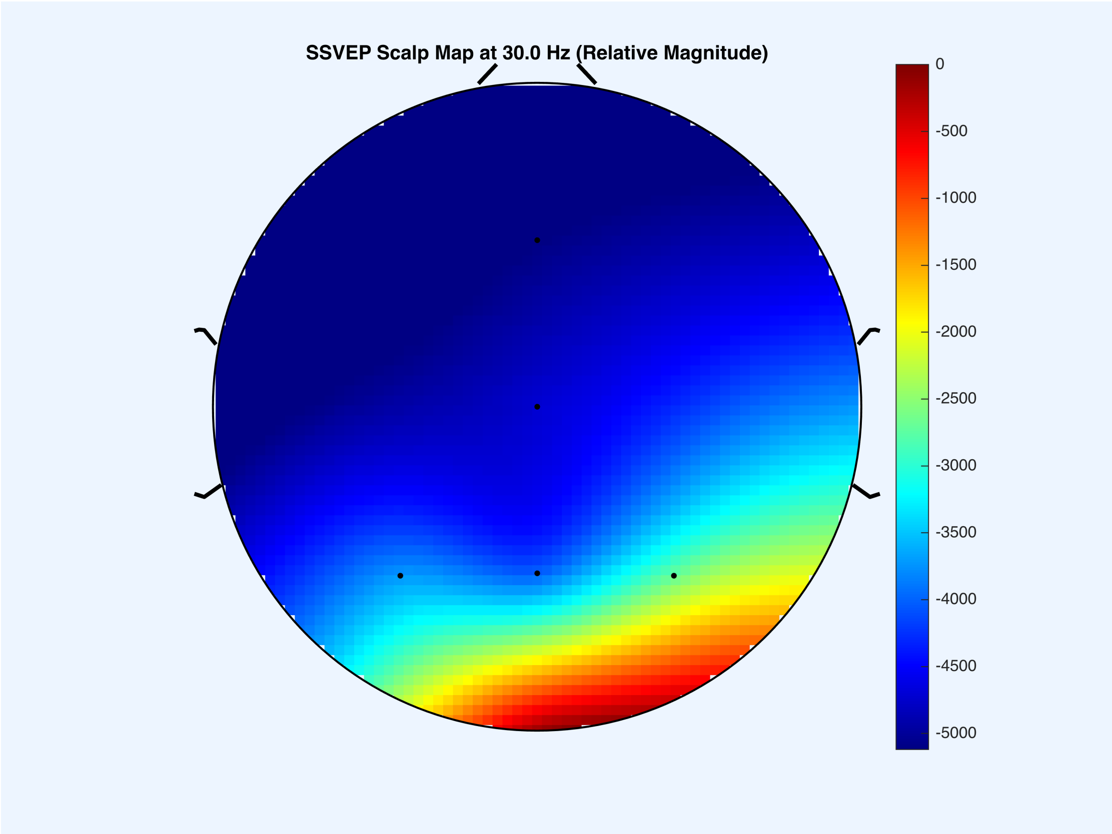
```
Warning: When plotting pvalues in totoplot, use option 'conv' to minimize extrapolation effects
```




```
lab8_exp1_12hz.mat: 12.0 Hz strongest = Oz (Magnitude 87536.38)
Warning: When plotting pvalues in totoplot, use option 'conv' to minimize extrapolation effects
```




```
lab8_exp1_20hz.mat: 20.0 Hz strongest = Oz (Magnitude 19415.39)
Warning: When plotting pvalues in totoplot, use option 'conv' to minimize extrapolation effects
```




```
lab8_exp1_30hz.mat: 30.0 Hz strongest = Oz (Magnitude 6890.32)
```

```
%% then indenitfy which condition produced the strongest response
avg_response = mean(magnitudes_all, 2); % avg across the channels

[~, idx_max] = max(avg_response);

fprintf('Strongest overall SSVEP response: %.1f Hz stimulus\n',
stim_freqs(idx_max));
```

```
Strongest overall SSVEP response: 12.0 Hz stimulus
```

Looking at all three of our stimuli, it is apparent that by far the strongest SSVEP magnitude occurred at Oz, with the 12 Hz stimulus managing to produce the largest response. I believe that we see this pattern for two reasons. First, It would seem that the Oz electrode lies directly over the primary visual cortex, which would then just generate the bulk of SSVEP activity. And then because these SSVEP effects originate in these early visual areas, it would then follow that the occipital electrodes would show the highest amplitude response.

Second, the lower-frequency flicker (that would be found in the 12Hz recording, usually around 10–15 Hz) is known to evoke these stronger steady-state responses due some different resonance properties of the visual system. From what I have read while completing this homework, the human visual cortex exhibits some increased sensitivity in the alpha band, meaning that 12 Hz is going to fall near this natural resonance range. As a result, we can see that the 12 Hz stimulation is able to drive a larger synchronized neural response than 20 Hz or 30 Hz, by far and away producing the highest spectral magnitude across all of our channels.

Question 3. Divide the strongest SSVEP case into 30s intervals and plot the normalized magnitude of the SSVEP signal as a function of the signal duration. At what duration can you reliably detect the presence of the SSVEP? (2 pts)

```
%still using the exp1 recordings
fs = 256;
remove_sec = 10;
Nremove = fs * remove_sec;

file = 'lab8_exp1_12hz.mat';
f0   = 12;  % should be the strongest SSVEP frequency
interval_sec = 30; % this is the length of each of our intervals
interval_len = fs * interval_sec;

S = load(file);
fn = fieldnames(S);
pre = S.(fn{1});

t   = pre(1,:);
EEG = pre(2:9,:);
EEG = EEG(:, Nremove+1:end);

for ch = 1:8
    EEG(ch,:) = detrend(EEG(ch,:), 'constant');
end

% this chunk will split into 30s intervals
L_total = size(EEG,2);
n_intervals = floor(L_total / interval_len);

magnitudes = zeros(n_intervals,1);

fprintf('Total usable samples: %d\n', L_total);
```

```
Total usable samples: 46062
```

```
fprintf('Interval length (samples): %d\n', interval_len);
```

```
Interval length (samples): 7680
```

```
fprintf('Number of intervals: %d\n\n', n_intervals);
```

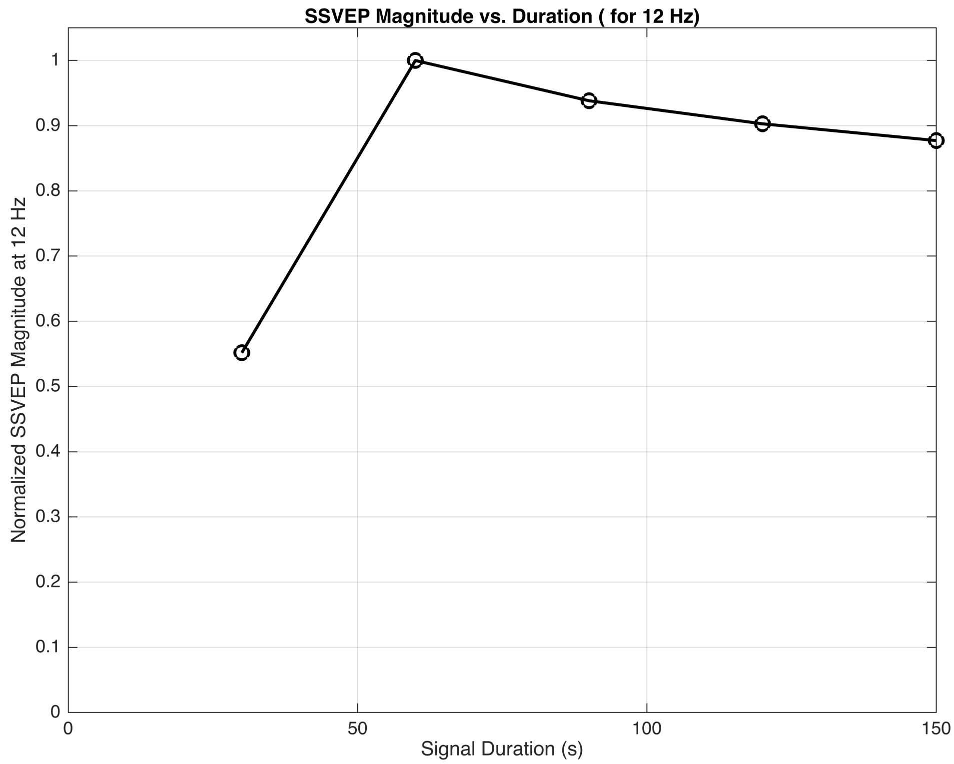
```
Number of intervals: 5
```

```

% compute SSVEP magnitudes
for k = 1:n_intervals
    idx_start = (k-1)*interval_len + 1;
    idx_end   = k*interval_len;
```

```
    segment = EEG(:, idx_start:idx_end);

    % fft for specified interval
    L = size(segment,2);
    df = fs / L;

    fft_seg = fft(segment, [], 2);
    mag_seg = abs(fft_seg);

    % bin for f0
    bin = round(f0 / df) + 1;

    %avg mag per channel
    magnitudes(k) = mean(mag_seg(:, bin));

    % sanity
    fprintf('Interval %d: bin %d -> %.3f Hz\n', k, bin, (bin-1)*df);
end
```

```
Interval 1: bin 361 -> 12.000 Hz
Interval 2: bin 361 -> 12.000 Hz
Interval 3: bin 361 -> 12.000 Hz
Interval 4: bin 361 -> 12.000 Hz
Interval 5: bin 361 -> 12.000 Hz
```

```
% then for the plot sake normalize
magnitudes_norm = magnitudes / max(magnitudes);

%plots
durations = (1:n_intervals) * interval_sec;

figure('Color','w');
plot(durations, magnitudes_norm, 'ko-', 'LineWidth',1.5, 'MarkerSize',7);
xlabel('Signal Duration (s)');
ylabel('Normalized SSVEP Magnitude at 12 Hz');
title('SSVEP Magnitude vs. Duration ( for 12 Hz)');
grid on;
xlim([0 durations(end)]);
ylim([0 1.05]);
```



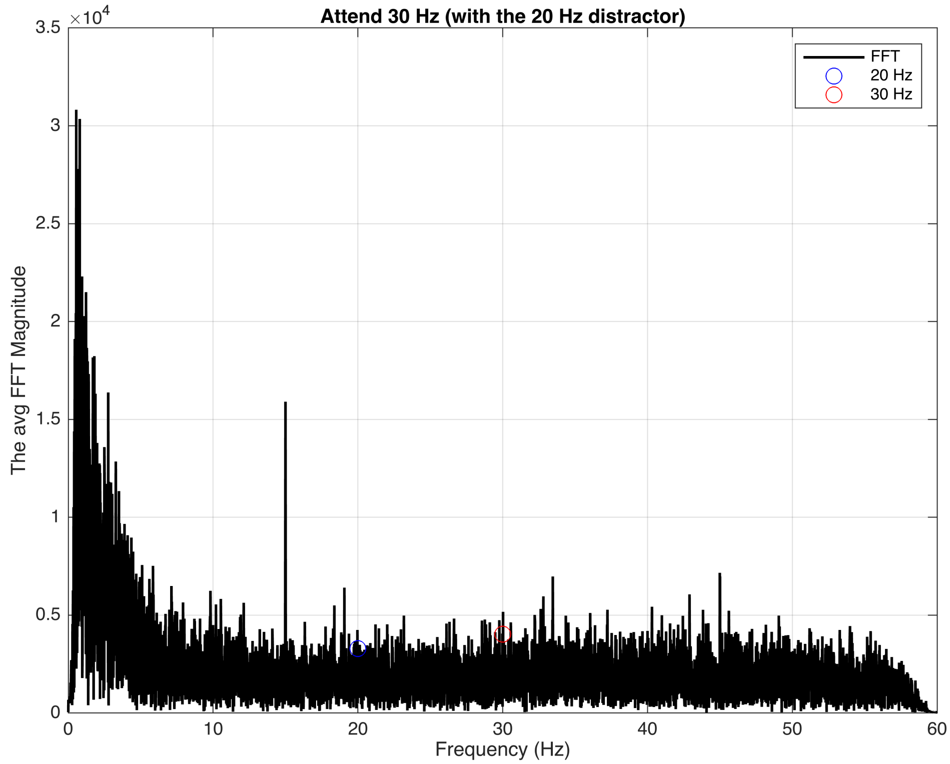
We can see that the normalized SSVEP magnitude increases substantially between the periods of 30s and 60s, which rises from about 0.55 normalized FFT magnitude to a value of 1. Looking at the output, by around 30 seconds, a clear 12 Hz peak is already present above the noise, but then we see that the response becomes unmistakably strong and stable by the 60s marker. After 60 seconds, the magnitude remains high (0.93 at 90 s, 0.90 at 120 s, 0.87 at 150 s), showing only minor fluctuations.

Therefore, I would feel confident in saying that the SSVEP is able to be reliably detected after about 30s of our recording data, and then is able to become fully robust by the 60s mark.

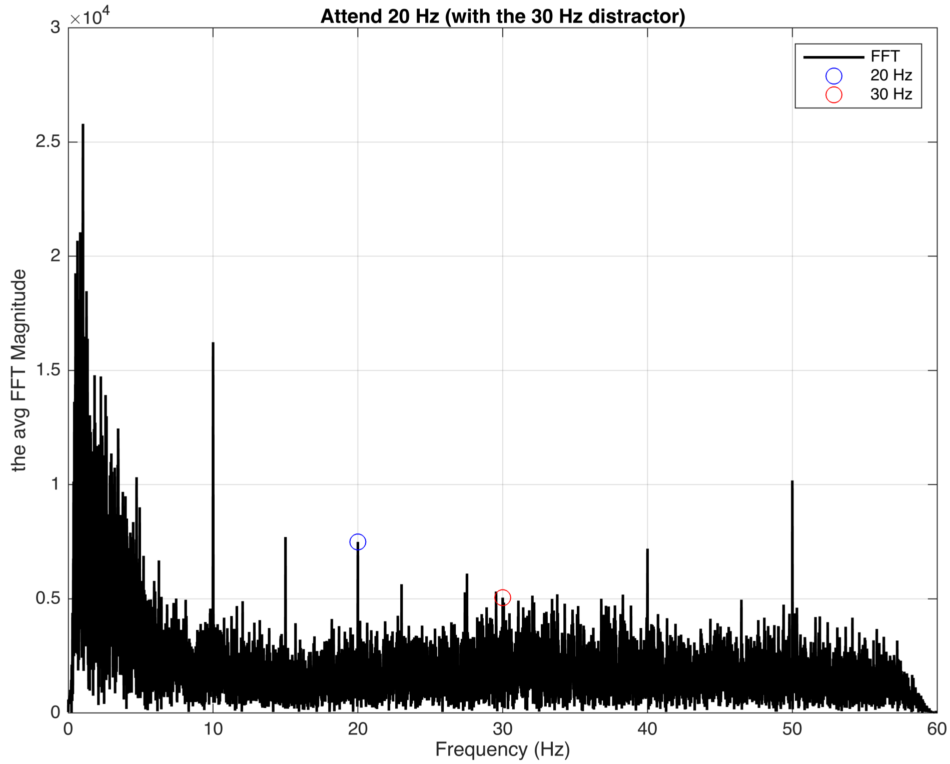
Question 4. From the data collected in the paradigm where you play 30Hz and 20Hz simultaneously and subject attends to only one of the two, plot FFT showing the SSVEP peaks for a) Attend 20Hz and distractor 30Hz and b) Attend 30Hz and distractor at 20Hz. What happens to 30Hz and 20 Hz peaks in FFT for condition a and b respectively? (2 pts)

```
fs = 256;
remove_sec_start = 10;
remove_sec_end   = 10;

file_raw = 'lab8_exp2_raw_trial3.mat';

% same recording as exp1, 1 = time and the rest are asc order
%electrodes we used.
S = load(file_raw);
fn = fieldnames(S);
```

```
Y  = S.(fn{1});

t   = Y(1,:);
EEG = Y(2:9,:);

% same 10s trimmed at the start
%our recording went long, meaning that we also want to trim the
%last 10s too
Nremove_start = fs * remove_sec_start;
Nremove_end   = fs * remove_sec_end;

t   = t(Nremove_start+1 : end-Nremove_end);
EEG = EEG(:, Nremove_start+1 : end-Nremove_end);
t   = t - t(1);
%same detrend
for ch = 1:8
    EEG(ch,:) = detrend(EEG(ch,:), 'constant');
end

% define the epochs of attention
%look right first (30hz) for 2 mins and then look left
attend30_idx = (t >= 0   & t < 120);
attend20_idx = (t >= 120 & t < 240);

EEG_att30 = EEG(:, attend30_idx);
EEG_att20 = EEG(:, attend20_idx);

%help function avg fft per channel
compute_avg_fft = @(X) deal( ...
    abs(mean(fft(X, [], 2), 1)), ...
    (0:size(X,2)-1) * (fs / size(X,2)));

[mag_att30, f_att30] = compute_avg_fft(EEG_att30);
[mag_att20, f_att20] = compute_avg_fft(EEG_att20);

% plotting the attend 30hz fft
figure('Color','w');
plot(f_att30, mag_att30, 'k', 'LineWidth', 1.3); hold on;
xlim([0 60]);
xlabel('Frequency (Hz)'); ylabel('The avg FFT Magnitude');
title('Attend 30 Hz (with the 20 Hz distractor)');
grid on;

[~, idx20_a] = min(abs(f_att30 - 20));
[~, idx30_a] = min(abs(f_att30 - 30));

plot(f_att30(idx20_a), mag_att30(idx20_a), 'bo', 'MarkerSize', 8);
plot(f_att30(idx30_a), mag_att30(idx30_a), 'ro', 'MarkerSize', 8);
legend('FFT','20 Hz','30 Hz');
```




```
% this is for the 20hz now
figure('Color','w');
plot(f_att20, mag_att20, 'k', 'LineWidth', 1.3); hold on;
xlim([0 60]);
xlabel('Frequency (Hz)'); ylabel('the avg FFT Magnitude');
title('Attend 20 Hz (with the 30 Hz distractor)');
grid on;

[~, idx20_b] = min(abs(f_att20 - 20));
[~, idx30_b] = min(abs(f_att20 - 30));

plot(f_att20(idx20_b), mag_att20(idx20_b), 'bo', 'MarkerSize', 8);
plot(f_att20(idx30_b), mag_att20(idx30_b), 'ro', 'MarkerSize', 8);
legend('FFT','20 Hz','30 Hz');
```




```
%this part is going to clean the ssvep mags then a localband snr

% Fz=1, Cz=2, Pz=3, P4=4, P3=5, PO8=6, PO7=7, Oz=8
occPar_idx = [3 4 5 6 7 8];

% new fft for the magnitude/SNR calculation

FFT_att30 = abs(fft(EEG_att30, [], 2));
FFT_att20 = abs(fft(EEG_att20, [], 2));

% find the magnitudes at the EXACT binsw e want
mag20_att30 = mean(FFT_att30(occPar_idx, idx20_a));
mag30_att30 = mean(FFT_att30(occPar_idx, idx30_a));

mag20_att20 = mean(FFT_att20(occPar_idx, idx20_b));
mag30_att20 = mean(FFT_att20(occPar_idx, idx30_b));

fprintf('\nRAW MAGNITUDES\n');
```

```
RAW MAGNITUDES
```

```
fprintf('Attend 30 Hz: 20 Hz = %.2f, 30 Hz = %.2f\n', mag20_att30,
mag30_att30);
```

```
Attend 30 Hz: 20 Hz = 4193.01, 30 Hz = 5788.68
```

```
fprintf('Attend 20 Hz: 20 Hz = %.2f, 30 Hz = %.2f\n', mag20_att20,
mag30_att20);
```

```
Attend 20 Hz: 20 Hz = 13144.08, 30 Hz = 8036.95
```

```

% then just a local band SNR function / exploring what happens to data

bw_sig = 0.5;
bw_nb  = [1 3];

compute_snr = @(fftmat, faxis, f0) ...
    ( ...
    mean(fftmat(:, faxis>=f0-bw_sig & faxis<=f0+bw_sig), 2) ./ ...
    mean(fftmat(:, ...
         (faxis>=f0-bw_nb(2) & faxis<=f0-bw_nb(1)) | ...
         (faxis>=f0+bw_nb(1) & faxis<=f0+bw_nb(2)) ...
     ), 2) ...
    );

snr20_att30 = mean(compute_snr(FFT_att30(occPar_idx,:), f_att30, 20));
snr30_att30 = mean(compute_snr(FFT_att30(occPar_idx,:), f_att30, 30));

snr20_att20 = mean(compute_snr(FFT_att20(occPar_idx,:), f_att20, 20));
snr30_att20 = mean(compute_snr(FFT_att20(occPar_idx,:), f_att20, 30));

fprintf('\n LOCAL-BAND SNR\n');
```

```
 LOCAL-BAND SNR
```

```
fprintf('Attend 30 Hz: SNR20 = %.3f, SNR30 = %.3f\n', snr20_att30,
snr30_att30);
```

```
Attend 30 Hz: SNR20 = 1.032, SNR30 = 1.103
```

```
fprintf('Attend 20 Hz: SNR20 = %.3f, SNR30 = %.3f\n', snr20_att20,
snr30_att20);
```

```
Attend 20 Hz: SNR20 = 1.072, SNR30 = 1.028
```

When our subject (me) was attending to the 30 Hz stimulus, I can see that the FFT plot was dominated by a very large “endogenous” 15 Hz peak (you can also make out its harmonic at 45 Hz), which initially made the stimulus-driven 30 Hz response that we were looking for rather difficult to identify by eye. However, one of the steps in making the report for this lab helped to explain this. Because a full black–white–black cycle requires two frames, the true visual stimulation frequency is actually half the nominal frame rate. That means the so-called “30 Hz” flicker is actually going to be driving the visual system at 15 Hz, which matches the dominant peak we observed almost exactly. In other words, what looked like a sort of strong intrinsic alpha rhythm at 15 Hz is actually going to be the SSVEP response to the 30 Hz (15 Hz effective) stimulus. Given this, this actually makes sense why we would see this 15 Hz component dominates the spectrum in that given interval.

Even so, in order to avoid relying solely on visual inspection of the FFT, I also included a quantitative analysis that showed that the extracted 30 Hz magnitude (5788.68) was still larger than the 20 Hz magnitude (4193.01) during this interval, indicating that the brain's response at the attended frequency still remained stronger even though the peak shifted to the true underlying stimulus frequency at 15 Hz.

In contrast, when I was observing the FFT for the 20 Hz interval, we can definitely see a clear peak at 10 Hz, which is again exactly half of the nominal 20 Hz frame rate. The numerical values confirm this enhancement as well, with the 20 Hz magnitude (13144.08) far exceeding the 30 Hz magnitude (8036.95). What is also worth mentioning is that we again see peaks around 10, 15, 27, 40, and 50 Hz, which from what I understand can reflect harmonics and intermodulation products generated because both flickers are present simultaneously and the visual system is nonlinear.

All in all, once we account for the fact that the true SSVEP frequency is half the frame rate, the spectrum makes entirely more sense. When this is combined with the quantitative magnitude analysis, this confirms the expected attention effect that we would expect to see, meaning that the attended frequency (in its true half-rate form) consistently dominates over the distractor frequency.